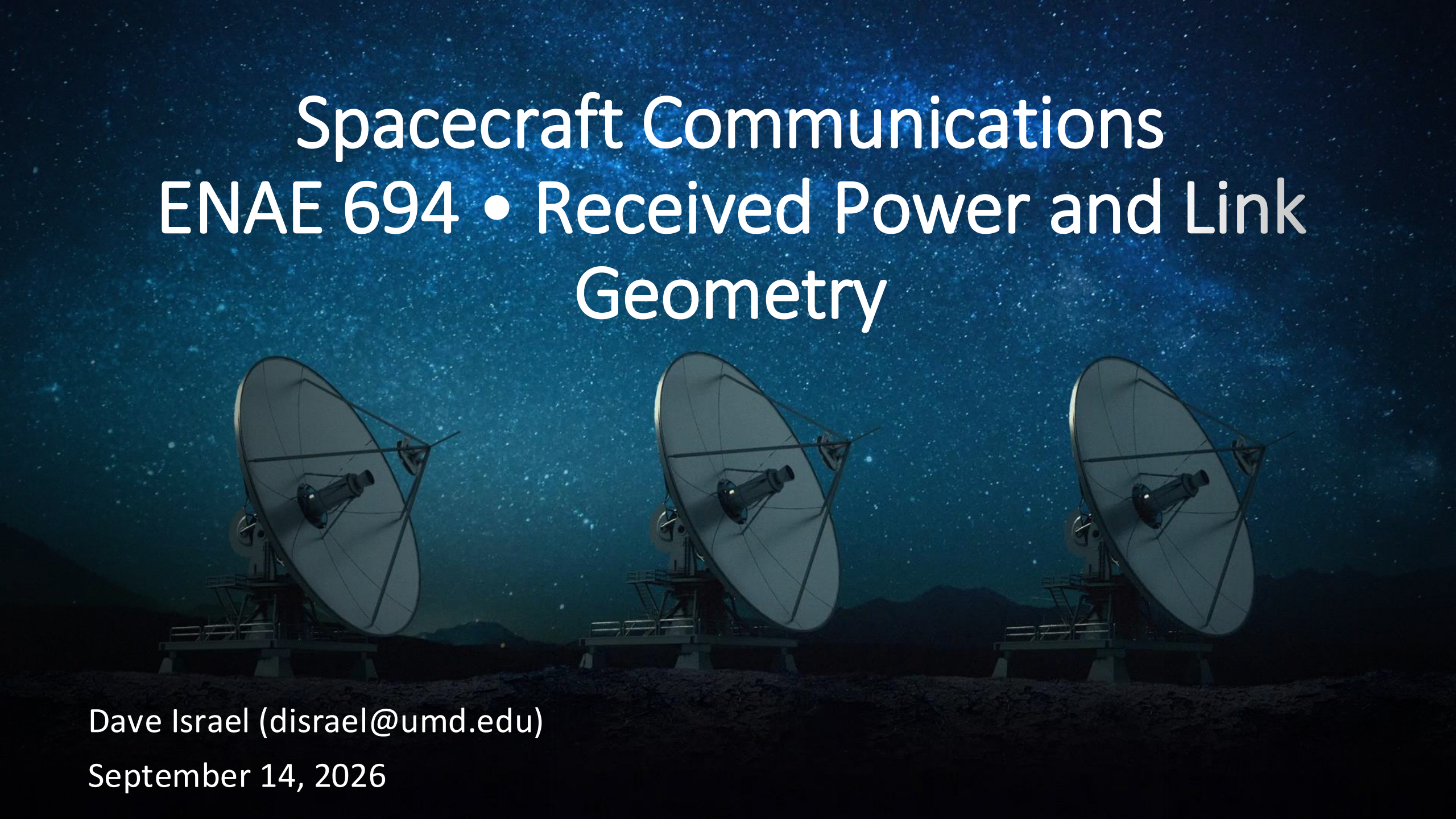
Spacecraft Communications
ENAE 694 • Received Power and Link Geometry



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The Physical Question

How much signal power reaches the receiver?

Tonight's Three Parts

1. Link Geometry and Slant Range
2. Received Power: Gains and Losses
3. The TERPSat Link Budget and Design Trades

One Link, Two Design Viewpoints

Spacecraft Team



Transmit Power and Antenna
Pointing and Orbit

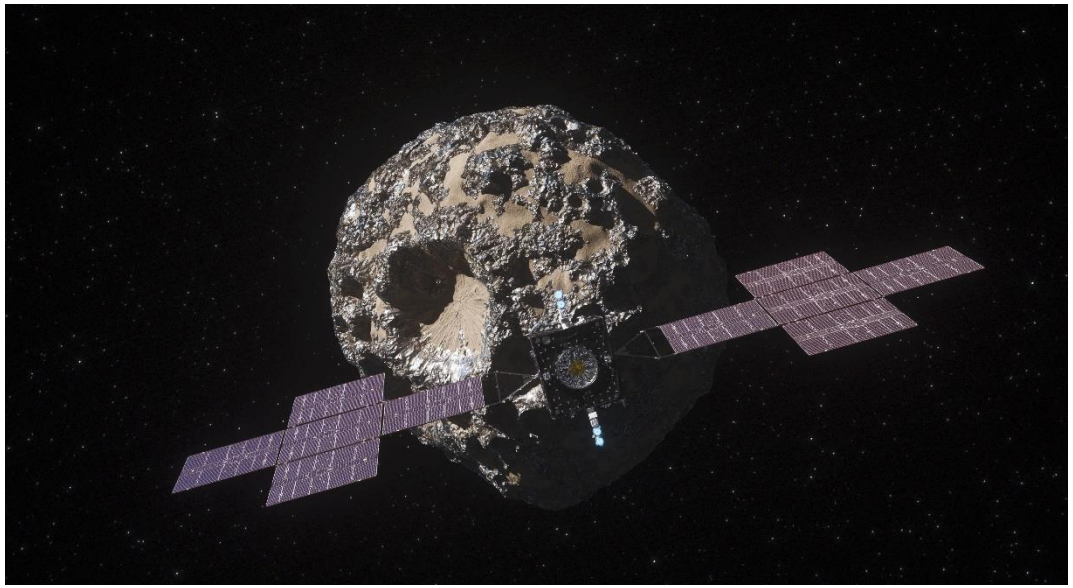
Network Provider



Station Antenna and Receiver
Supported Frequencies and Scheduling



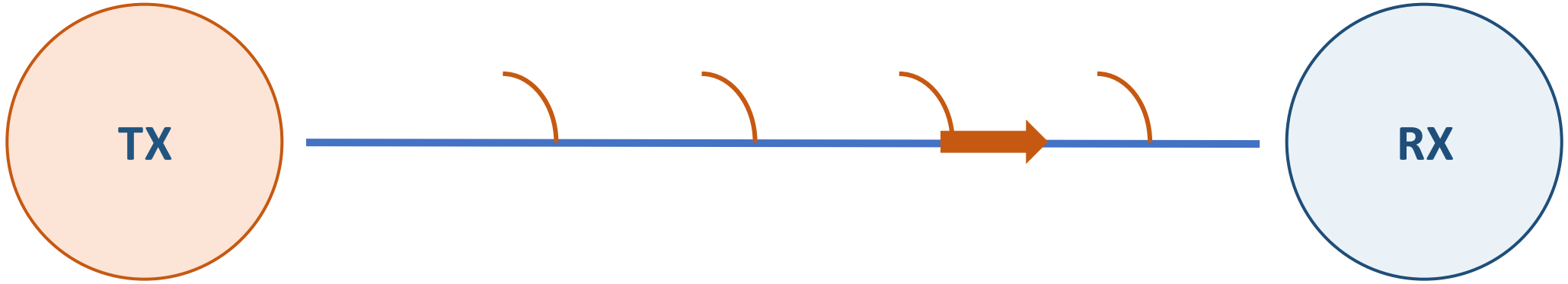






One Direction at a Time

one transmitter → one receiver → one direction



The reciprocal link needs its own budget. Transmitter and receiver parameters can differ.

Received Power Is Only a Start

Tonight we calculate carrier power at the receiver input

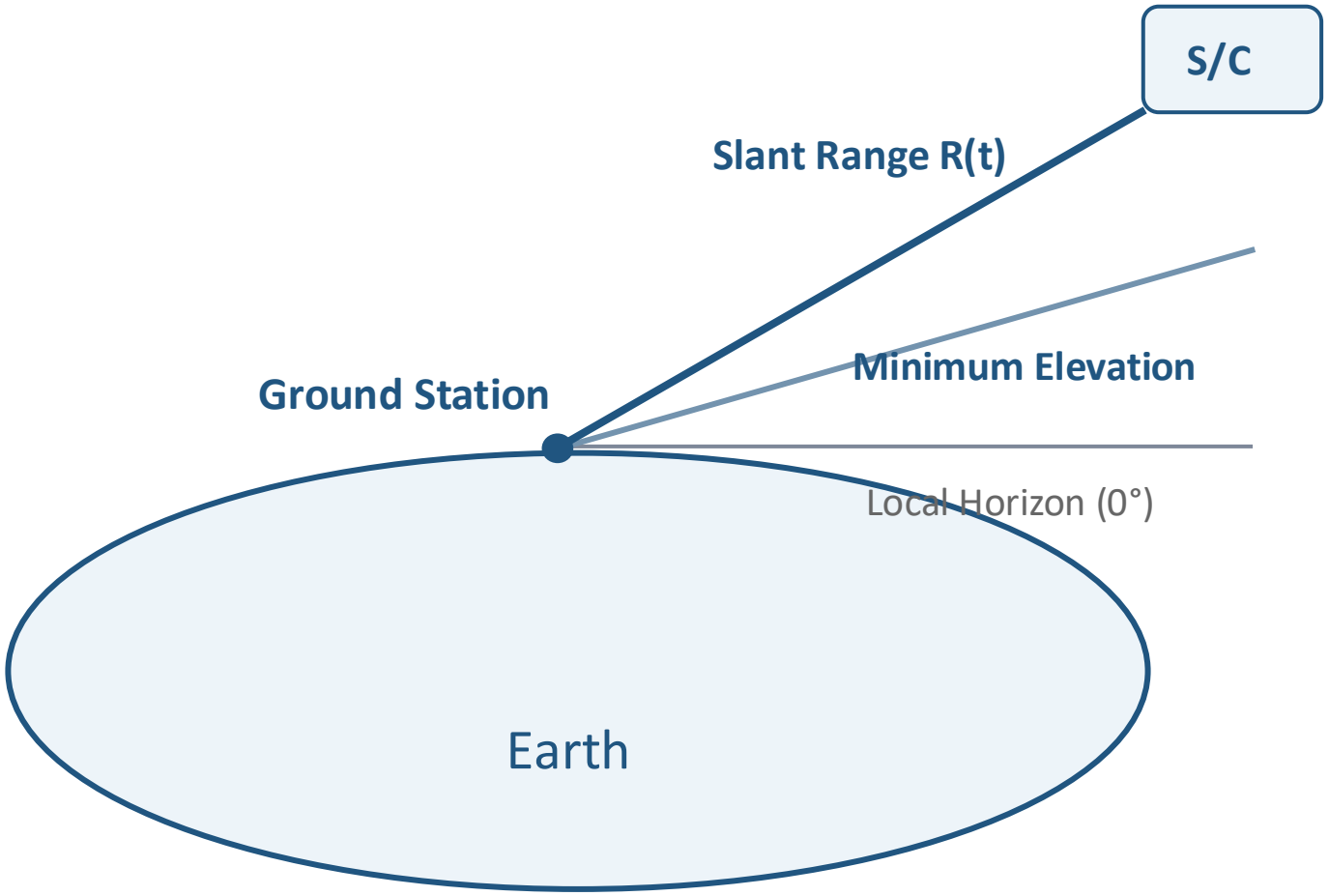
We do not yet know whether the receiver can recover the data

That requires noise, waveform performance, and margin

A respectable number in dBW is not yet a working communications service

When Does a Path Exist?

1. **Earth, terrain, or structures can block line of sight**
2. **Service begins above e_{min}**
3. **Slant range R(t) changes throughout the pass**
4. **Design to the worst useful geometry—not merely visibility**



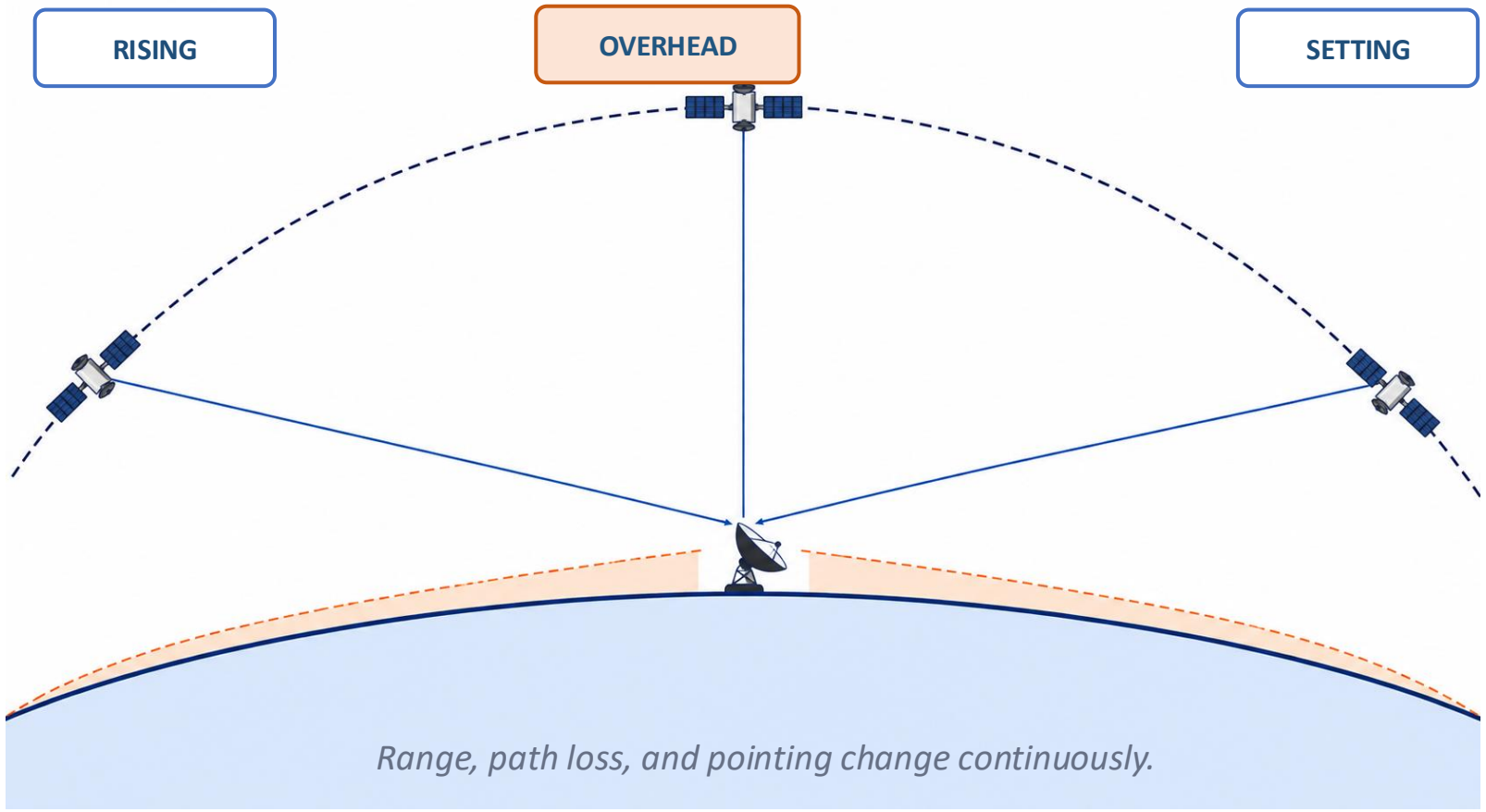
Higher Elevation Mask: Shorter Maximum Range, Less Contact Time

A Pass Is Not One Range

AOS
long range
weakest power

MAXIMUM ELEVATION
shortest range
strongest power

LOS
long range
weakest power



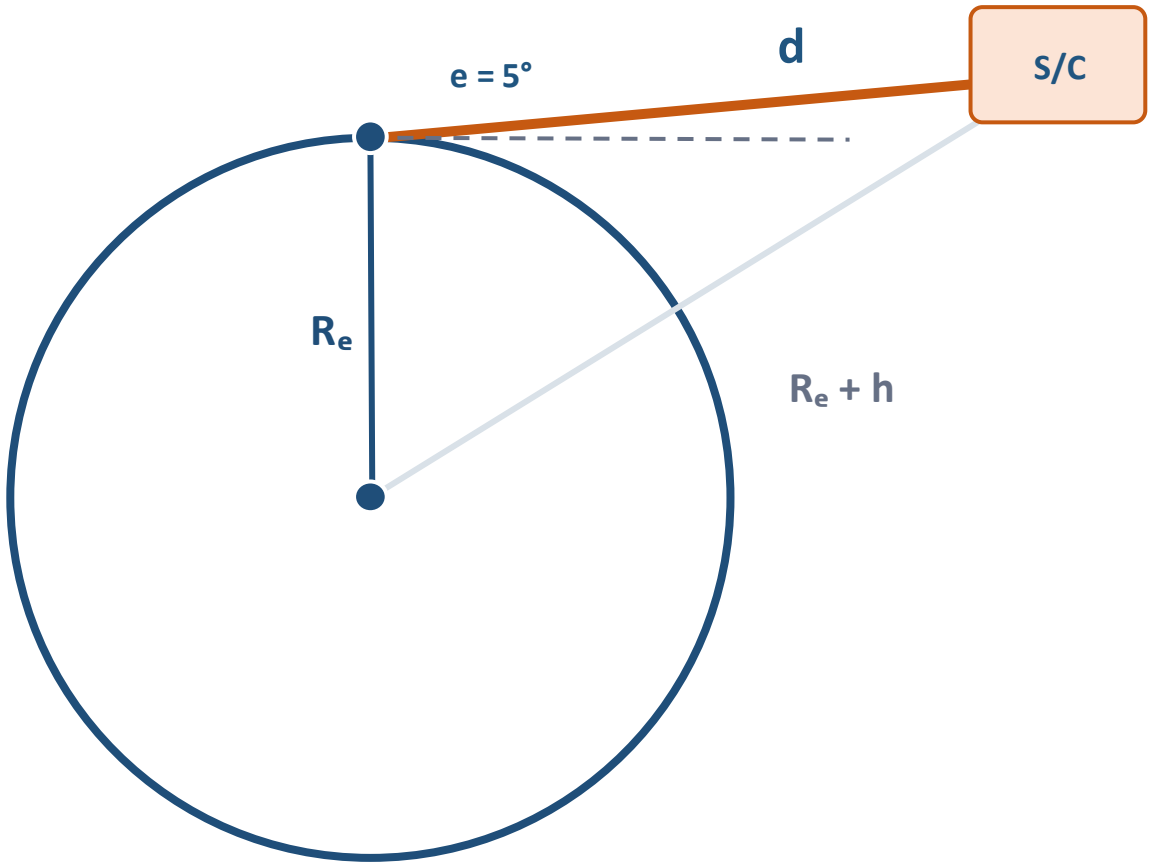
AOS = Acquisition of Signal LOS = Loss of Signal

TERPSat Link Geometry

EARTH RADIUS
6,378 km

ORBIT ALTITUDE
705 km

MINIMUM ELEVATION
5°



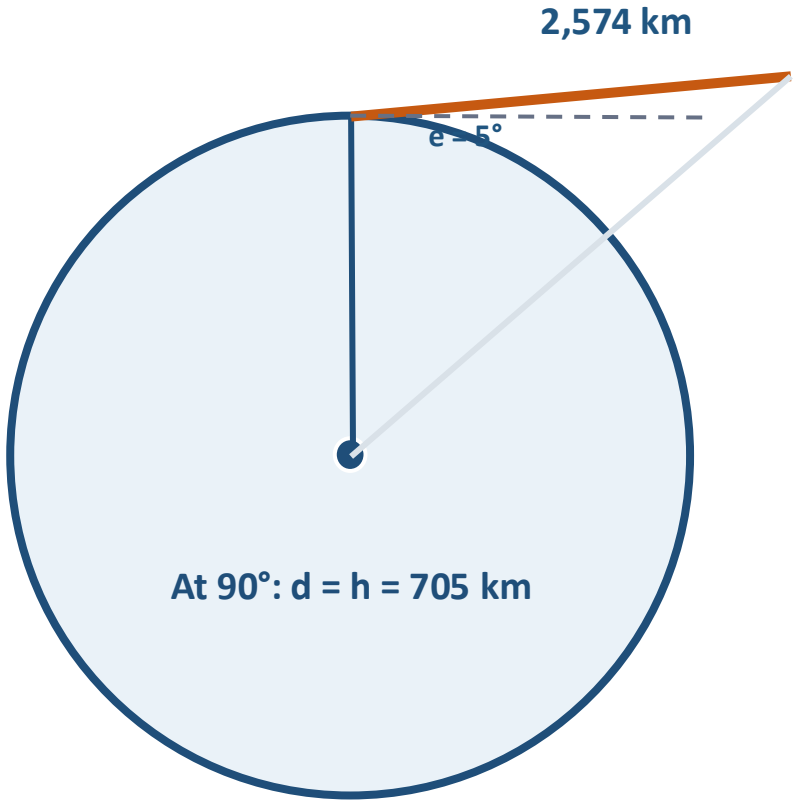
Calculating Slant Range

$$d = \sqrt{(R_e + h)^2 - R_e^2 \cos^2 e} - R_e \sin e$$

$R_e = 6{,}378$ km • $h = 705$ km • $e = 5°$

$d \approx 2{,}574$ km

Geometry alone creates a 3.65:1 range swing.



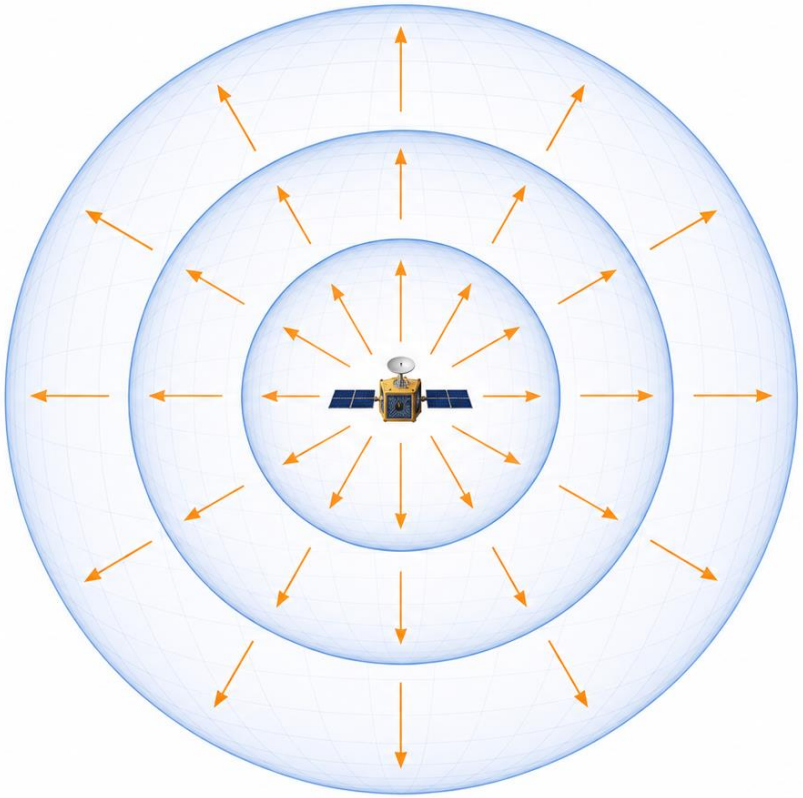
Why Power Falls with Range

Same transmitted power.
Larger spherical area.

$$\text{AREA} = 4\pi R^2$$

$$\text{FLUX} \propto 1/R^2$$

R → 2R
area ×4 • flux ÷4
$\Delta[P_r] = -6$ dB



What Is a Decibel?

Power ratio in dB = $10 \log_{10}(P_2 / P_1)$

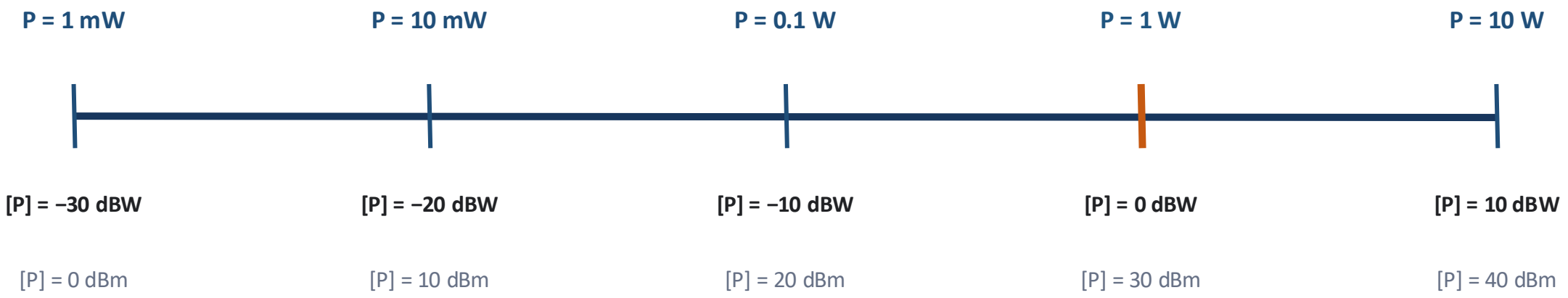
[P] in dBW = $10 \log_{10}(P / 1\ \text{W})$

P is linear power; [P] is its decibel representation.

Try it: 1 W → 10 W → 100 W. What about 2 W?

Linear Power and Decibel Power

Course notation: P is linear power; [P] is that power expressed in dB.



Every +10 dB step multiplies power by 10. Gains add; losses subtract.

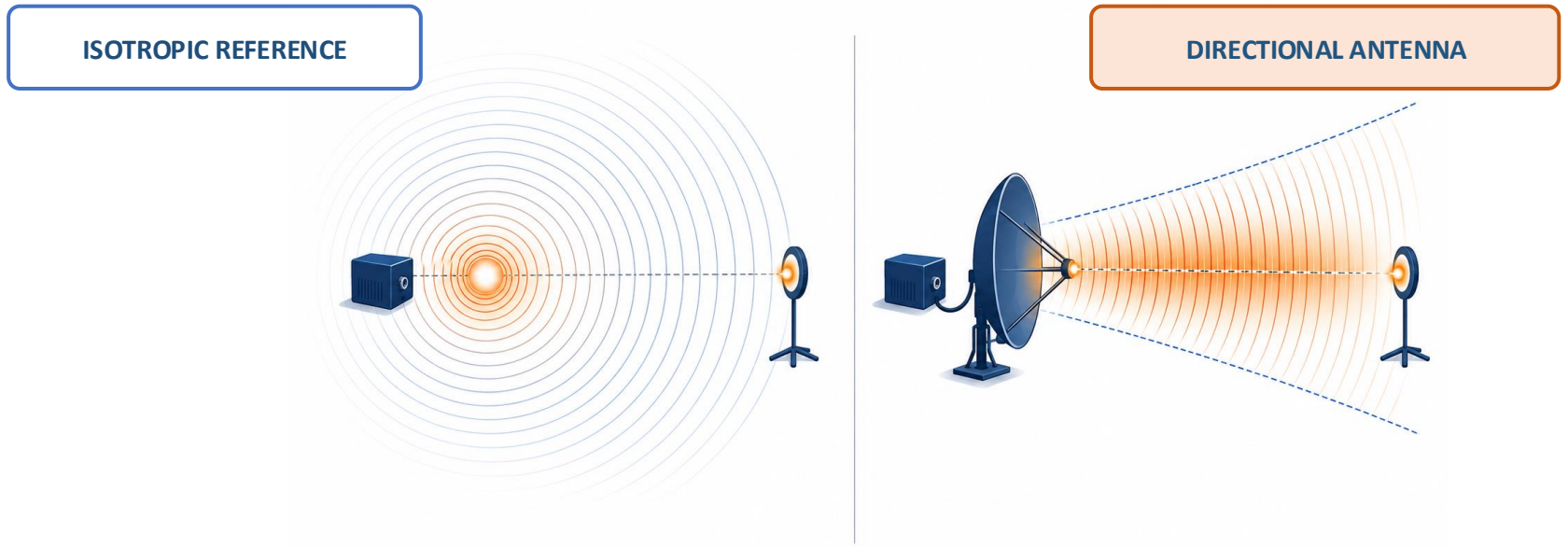
Useful Decibel Relationships

Predict the sign and scale before touching the calculator.

CHANGE	PHYSICAL EFFECT	RECEIVED POWER
POWER ×2	twice the watts	$\Delta[P_r] = +3$ dB
POWER ×10	ten times the watts	$\Delta[P_r] = +10$ dB
RANGE ×2	four times the sphere area	$\Delta[P_r] = -6$ dB
RANGE ×10	one hundred times the area	$\Delta[P_r] = -20$ dB

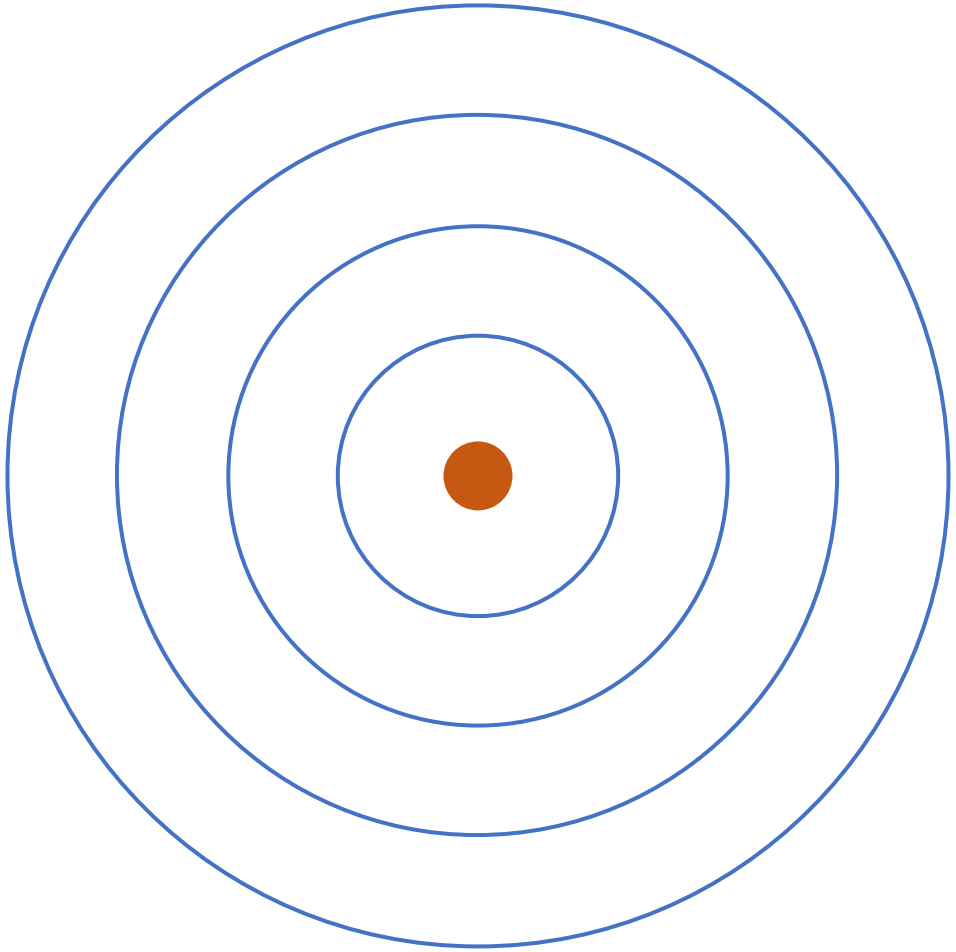
Equivalent Isotropically Radiated Power

$$[EIRP] = [P_t] - [L_t] + [G_t]$$

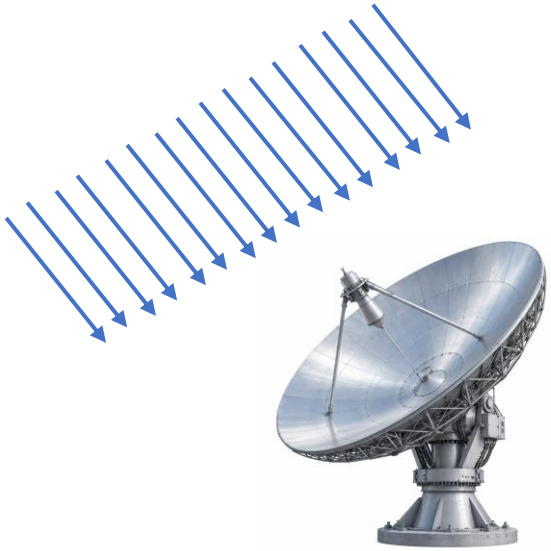


Equivalent on-axis intensity—not a new power amplifier.

Collecting the Power



$S = EIRP / (4\pi R^2)$, in W/m^2



Only the field crossing the effective collecting area is captured.

$P_r = S\ A_e$ (linear quantities)

Calculating Free-Space Path Loss

$[L_p] = 20 \log_{10}(4\pi R / \lambda)$

In km and GHz: $[L_p] = 92.45 + 20 \log_{10}(R) + 20 \log_{10}(f)$

For TERPSat: R = 2574 km and f = 8.4 GHz.

$[L_p]$ is a positive loss; subtract it in the power ledger.

The Received-Power Equation

Spacecraft: [EIRP]

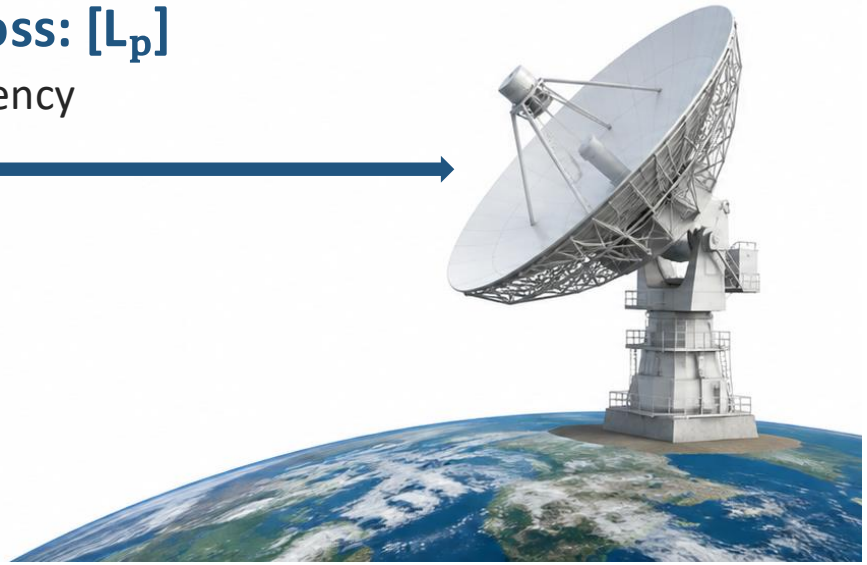
Transmitter + Feed + Antenna



Receive Antenna: [G_r]

Free-Space Path Loss: [L_p]

Range and Frequency



$[EIRP] = [P_t] - [L_t] + [G_t]$

Transmit Power − Feed Loss + Antenna Gain

$[P_r] = [EIRP] - [L_p] + [G_r] - [L_{other}]$

Power: dBW Antenna Gain: dBi Losses: dB

Other losses exclude the transmit feed loss already included in EIRP.

Work the TERPSat Link

At 5° elevation: d = 2,574 km

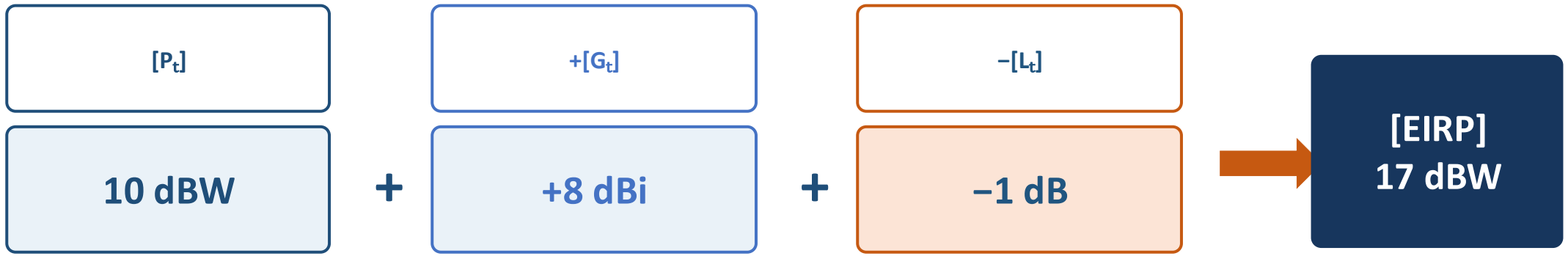
Frequency: f = 8.4 GHz

P_t = 10 W • $[G_t]$ = 8 dBi • $[L_t]$ = 1 dB

$[G_r]$ = 45 dBi • $[L_{other}]$ = 2 dB

Calculate [EIRP], $[L_p]$, and $[P_r]$.

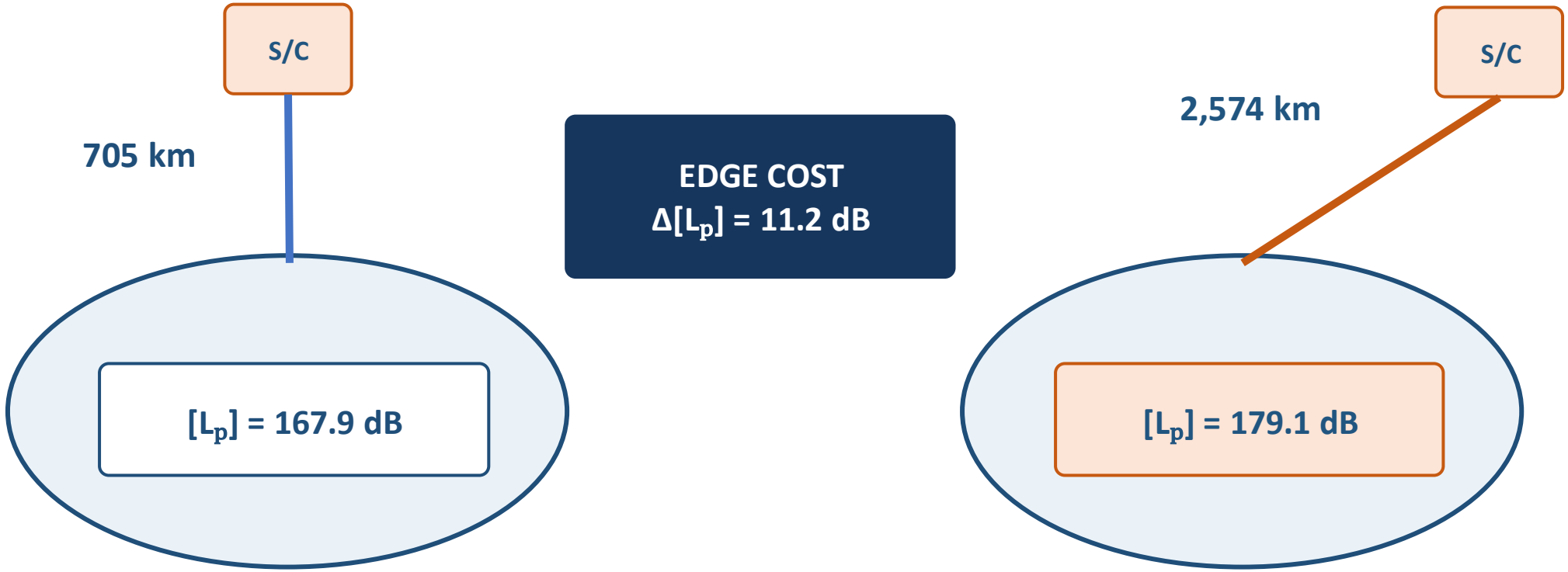
Step 1: EIRP



The spacecraft still transmits 10 W. Antenna gain changes the on-axis equivalent.

Step 2: Free-Space Path Loss

At one frequency, free-space loss changes only because range changes.



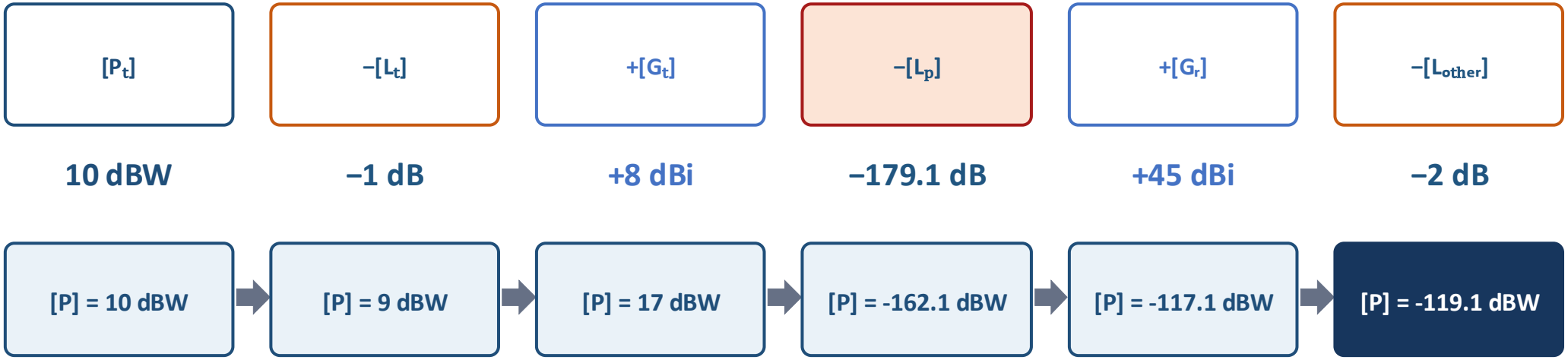
Step 3: Received Power



Does the link work? We still need noise, required performance, and margin.

Check the Complete Ledger

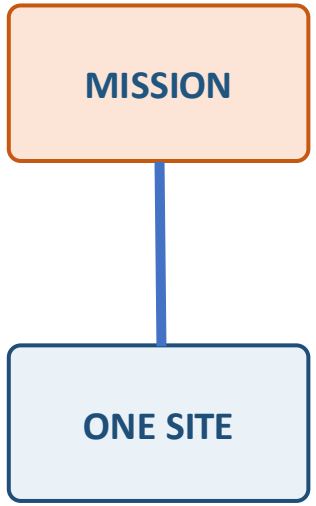
Follow the reference point. Write every gain and loss with its sign.



The path loss is enormous; antenna gains make the link plausible.

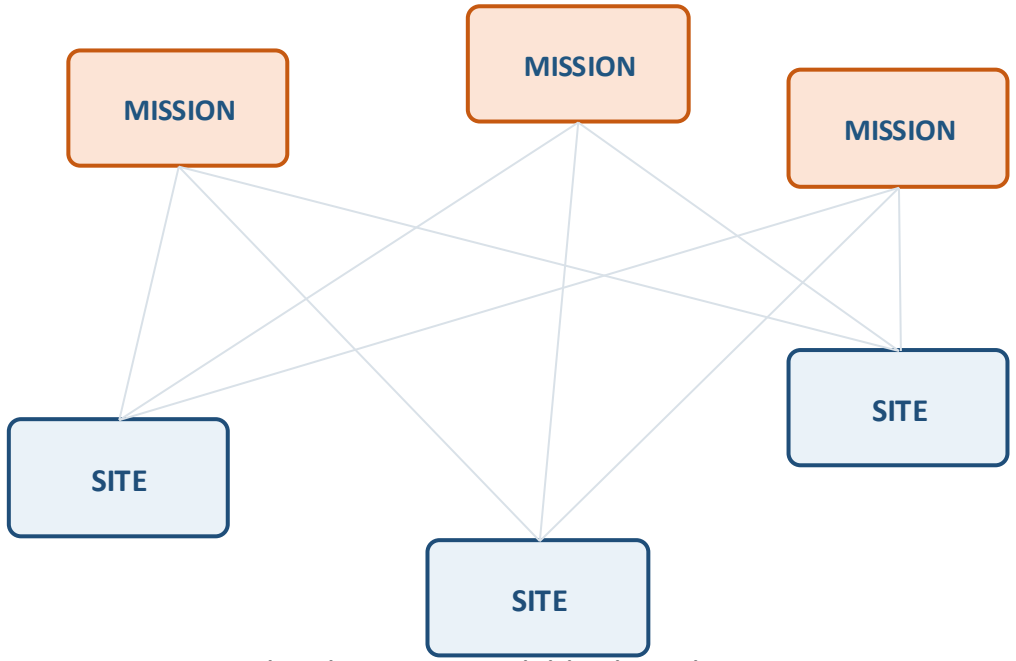
Mission and Provider Trades

MISSION-OWNED



Direct control • tailored interface • dedicated capacity
Capital and operations burden • limited geography

SHARED PROVIDER



Geographic diversity • scalable shared assets
Scheduling, interfaces, and service dependence

Predict Before Calculating

What happens to received power if we double transmit power?

What happens if we double range instead?

What do we gain—and lose—by raising the elevation mask?

Does adding a ground station necessarily strengthen this link?

References and Further Reading

CITATIONS & REFERENCES

- ITU-R P.525-4, Calculation of Free-Space Attenuation (2019)
- JPL 810-005, DSN Telecommunications Link Design Handbook
- CCSDS 401.0-B-29, RF and Modulation Systems—Part 1 (2019)
- Friis, “A Note on a Simple Transmission Formula,” Proc. IRE 34(5), 1946

OPTIONAL FURTHER READING

- Cheung, “The Role of Margin in Link Design and Optimization” (2015)
- NASA, Small Spacecraft Technology State of the Art: Communications (2026)
- Pratt, Bostian & Allnutt, Satellite Communications, 2nd ed.
- Vallado, Fundamentals of Astrodynamics and Applications, 5th ed.

Selected References and Optional Further Reading